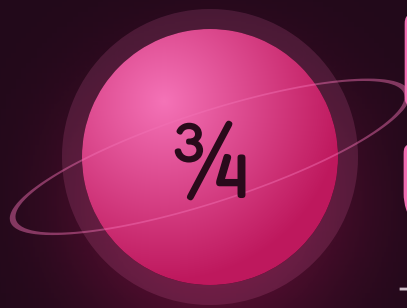




Fraction Universe

The mission where one number has many names.

THE BIG QUESTION

How can two different-looking fractions be exactly the same number?

HANDS-ON PROJECT

Fold the Universe

Build a paper number line from 0 to 1 and land every fraction you meet on it.

FLUENCY GAME

Closest to One

Draw cards, build a fraction, get nearest to 1 without going over.



AXIOM • MISSION COMMANDER, NINE TOURS

“Brock — six weeks. A fraction is one number, not two stacked up. Get that on day one and the rest is bookkeeping.”

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

This is not a test. Nothing here counts for a grade. Six weeks is a long mission — every strand you already own takes days off it.

STRAND A NUMBER LINE

A1 Mark $\frac{3}{4}$ on this line.

0 _____ 1

A2 How many thirds are in 2 wholes?

.....

STRAND B EQUIVALENCE

B1 $\frac{1}{2} = \frac{\quad}{8}$

.....

B2 $\frac{3}{4} = \frac{\quad}{12}$

.....

STRAND C COMPARING

C1 Circle the bigger: $\frac{2}{3}$ $\frac{3}{5}$

.....

C2 Order smallest first: $\frac{1}{2}$, $\frac{3}{4}$, $\frac{2}{5}$

.....

STRAND D ADD & SUBTRACT

D1 $\frac{3}{8} + \frac{2}{8} = \frac{\quad}{\quad}$

.....

D2 $\frac{1}{2} + \frac{1}{4} = \frac{\quad}{\quad}$

.....

STRAND E SIMPLEST FORM

E1 $\frac{10}{15} = \frac{\quad}{\quad}$

.....

E2 $\frac{9}{12} = \frac{\quad}{\quad}$

.....

STRAND F OF A SET

F1 $\frac{2}{5}$ of 30 = _____

.....

F2 $\frac{3}{4}$ of a number is 18. What is the number?

.....

Skip Chart

Score the pre-assessment, then cross weeks off the six-week calendar.

2 / 2 → SKIP

1 / 2 → COMPRESS

0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A • Number line	A1 A2	Week 1 • lesson 1.1	___ / 2	Day 1.1. Keep the folding project regardless — it earns its time.
B • Equivalence	B1 B2	Week 1 • 1.2 and all of Week 2	___ / 2	Week 2 days 2.1–2.3. Keep 2.4 — fraction families are new to most.
C • Comparing	C1 C2	Week 1 • 1.3 and Week 3	___ / 2	Week 3 days 3.1–3.3.
D • Add & subtract	D1 D2	Week 1 • 1.4 and Week 4	___ / 2	Week 4 days 4.1–4.2. Never skip 4.3 — both denominators changing is the real skill.
E • Simplest form	E1 E2	Week 2 • lesson 2.2	___ / 2	Day 2.2 only.
F • Of a set	F1 F2	Week 5	___ / 2	Days 5.1–5.2. Keep 5.3, working backwards.

Where the saved time goes

Into the Challenge tier and the folded number line, or into an early start on Mission 05 — decimals are the same idea in different clothes, so momentum here is worth keeping. Write the reallocation:

.....

Revised Mission 04 plan

WEEK 1 ☐ full ☐ compressed ☐ skipWEEK 2 ☐ full ☐ compressed ☐ skipWEEK 3 ☐ full ☐ compressed ☐ skipWEEK 4 ☐ full ☐ compressed ☐ skipWEEK 5 ☐ full ☐ compressed ☐ skipWEEK 6 ☐ full ☐ compressed ☐ skip

One Whole, Cut Up

The bottom number cuts. The top number counts.

Warm-Up 6 ITEMS • 5 MINUTES

halves in 1 whole

quarters in 1 whole

eighths in 1 whole

$1/2 = ?/4$

$1/2 = ?/8$

$3/3 =$ how many wholes

THE SAME BAR, CUT FOUR WAYS

1 whole

halves

quarters

eighths

The shaded part never moved. Write what each row calls it: _____, _____, _____, _____

Core

1. $1/4 + 1/4 + 1/4 = \underline{\quad} / 4$

2. How many thirds in 2 wholes?

3. $5/5$ is how many wholes?

4. $7/4 = \underline{\quad}$ whole and $\underline{\quad}$ quarters

5. Mark $1/4$, $1/2$ and $3/4$ on this line. Then mark $5/8$ and say which two it sits between.

0 _____ 1

★ Challenge NOT GRADED

C1. If you cut 0 to 1 into fifths, how many marks are there between 0 and 1?

.....

C2. $3/8$ and $5/8$ — how many eighths apart are they on the line?

.....

C3. Which is a bigger piece, $1/8$ or $1/5$? Explain why the bigger bottom number makes the smaller piece.



AXIOM: Say "three eighths", never "three over eight". The words tell you what it is; the symbol only tells you how to write it.

Cut every piece in two and nothing about the amount changes.

Warm-Up

$$\frac{1}{2} = \frac{?}{6}$$

$$\frac{1}{3} = \frac{?}{6}$$

$$\frac{2}{3} = \frac{?}{6}$$

$$\frac{1}{4} = \frac{?}{8}$$

$$\frac{3}{4} = \frac{?}{8}$$

$$\frac{1}{5} = \frac{?}{10}$$

THE MOVE

$$\frac{3}{4} = \frac{9}{12}$$

bottom: $4 \times 3 = 12$

top: $3 \times 3 = 9$

Whatever you do to the bottom you must do to the top, because you are cutting every piece the same way — not adding anything.

 Core WRITE WHAT THE BOTTOM WAS MULTIPLIED OR DIVIDED BY

1. $3/4 = \underline{\quad\quad\quad} / 12$ bottom $\times \underline{\quad\quad}$

2. $2/5 = \underline{\hspace{1cm}} / 20$ bottom $\times \underline{\hspace{1cm}}$

3. $6/8 = \underline{\quad} / 4$ bottom $\div \underline{\quad}$

4. $10/15$ simplified both $\div$ _____

5. $\frac{9}{12}$ simplified both $\div$ _____

★ Challenge

C1. $8/12 = \underline{\hspace{1cm}} / 3$. What did you divide both numbers by?

C2. List every fraction equal to $\frac{1}{2}$ with a denominator under 20. How many are there?

C3. Draw a picture that proves $\frac{2}{3} = \frac{8}{12}$ without doing any multiplying.

[illegible]

Which Is Bigger

You can't compare pieces of different sizes. Make them match, then count.

✦ Warm-Up CIRCLE THE BIGGER ONE

1/2 1/4

1/3 1/5

3/4 1/4

2/5 3/5

5/6 5/8

7/10 3/10

WORKED EXAMPLE • 2/3 AGAINST 3/5

 $\frac{2}{3} =$
 $\frac{10}{15}$
 $\frac{3}{5} =$
 $\frac{9}{15}$

Ten fifteenths beats nine fifteenths. Finding a common denominator is just making the pieces the same size.

◆ Core REWRITE BOTH, THEN CIRCLE THE BIGGER

1. $\frac{2}{3}$ vs $\frac{3}{5}$ in fifteenths: _____ and _____

2. $\frac{3}{4}$ vs $\frac{5}{8}$ in eighths: _____ and _____

3. $\frac{5}{6}$ vs $\frac{7}{9}$ in eighteenths: _____ and _____

4. $\frac{2}{3}$ vs $\frac{3}{4}$ in twelfths: _____ and _____

5. Closer to 1: $\frac{7}{8}$ or $\frac{5}{6}$? Compare the gaps, not the fractions:

★ Challenge

C1. Order $\frac{3}{5}$, $\frac{2}{3}$ and $\frac{7}{10}$ smallest first. Use thirtieths.

.....

C2. Find the fraction exactly halfway between $\frac{1}{2}$ and $\frac{3}{4}$. Write it in eighths.

.....

C3. Without any calculation, is $\frac{7}{15}$ more or less than a half? How can you tell at a glance?

.....

Adding Needs Matching Pieces

Add the tops. The bottom is the unit — it never gets added.

Warm-Up BOTTOMS ALREADY MATCH

$1/4 + 1/4$

$2/5 + 1/5$

$3/8 + 2/8$

$5/6 - 2/6$

$7/10 - 4/10$

$3/4 + 1/4$

WORKED EXAMPLE • $1/2 + 1/4$

$1/2 + 1/4$

$1/2$

$1/4$

$2/4 + 1/4$

Once every piece is a quarter you can count them: three. $1/2 + 1/4 = 3/4$ — and the bottom stayed 4 the whole time.

Core REWRITE FIRST, THEN ADD

$1. 1/2 + 1/4$

$2. 2/3 + 1/6$

$3. 3/4 - 1/8$

$4. 5/6 - 1/3$

$5. 1/2 + 1/3$ Neither one fits the other. What size do both become?

★ Challenge

C1. $3/4 + 2/3 =$ _____. The answer is bigger than 1 — say how you know before you work it out.

.....

C2. $2 - 3/5 =$ _____. Turn the 2 into fifths first.

.....

C3. Someone writes $1/2 + 1/4 = 2/6$. What did they do, and why is it wrong? Use the bar picture in your answer.

.....

Fold the Universe

One strip of paper, folded again and again. Every fold is a denominator.

The build • four folds

Fold 1. In half. Mark $\frac{1}{2}$ on the crease.

Fold 2. In half again. Mark $\frac{1}{4}$ and $\frac{3}{4}$.

Fold 3. Again. Mark every eighth.

Fold 4. Again. Sixteenths — and now look at which marks landed on top of each other.

Which fractions share a crease? List three pairs:

★ Fraction of a set DIVIDE BY THE BOTTOM, MULTIPLY BY THE TOP

$\frac{1}{4}$ of 20

$\frac{2}{5}$ of 30

$\frac{3}{8}$ of 40

$\frac{2}{3}$ of 45

$\frac{5}{6}$ of 42

$\frac{3}{4}$ of a number is 18.
The number?

Closest to One • two players, one deck

Strip the face cards. Each player draws two cards and builds a fraction from them — you choose which card goes on top. Closest to 1 *without going over* wins the round. Going over scores nothing, so a bold fraction is a risk.

My best fraction this game:

How far from 1 it was:

The real question

You draw a 5 and a 6. Which way round should you build the fraction, and why? Now say what the best possible two-card hand in the whole deck would be.

.....
.....

The Rest of the Mission

Five weeks left — the longest stretch of the year. Same three-tier shape every day.

Week 2 · Equivalence

THE MOST USEFUL MOVE IN THE MISSION

2.1 Mon

Multiply top and bottom. The same amount in different clothes.

2.2 Tue

Simplest form. Divide until nothing divides both.

2.3 Wed

Missing numerators:
 $3/4 = ?/12$.

2.4 Thu

Fraction families.
Every fraction equal to $1/2$.

Fri

Equivalence wall of paper strips.

Week 3 · Comparing & Ordering

MID-UNIT QUIZ FRIDAY • 85% TO ADVANCE

3.1 Mon

Common denominators. Make the pieces match.

3.2 Tue

Benchmarks. More or less than a half?

3.3 Wed

Ordering four at once, with a defence.

3.4 Thu

Closer to 1 or 0?
Compare the gaps.

Fri

Mid-unit quiz, 8 items, Weeks 1-3.

Week 4 · Adding & Subtracting

THE GRADE-FIVE CEILING LIVES HERE

4.1 Mon

Like denominators. The bottom does not change.

4.2 Tue

Unlike denominators. Convert one, then add.

4.3 Wed

Both need changing:
 $1/2 + 1/3$.

4.4 Thu

Subtracting from a whole: $2 - 3/5$.

Fri

Closest to One, played for real.

Week 5 · Fraction of a Set

FRACTIONS BECOME OPERATORS

5.1 Mon

Split, then take. The two numbers do different jobs.

5.2 Tue

Bigger sets: $5/6$ of 42.

5.3 Wed

Working backwards:
 $3/4$ of it is 18.

5.4 Thu

Fractions of time and money.

Fri

Fold the Universe finished.

Week 6 · Proof & Project

UNIT TEST FRIDAY • BADGE AWARDED

6.1 Mon

Prove $2/3 = 8/12$ with a drawing, not arithmetic.

6.2 Tue

Number line defence, out loud.

6.3 Wed

Mixed problems from all five weeks.

Thu

Error journal sweep.

Fri

Mission 04 test: 12 items + one explanation.

MATERIALS

Strips of paper for folding ·
fraction tiles or a printed set ·
deck of cards · coloured pencils ·
the error journal

WEEKLY CHECK

Five items every Friday, mixed tiers. Below $4/5$ means Monday is a reteach, not new material.

THE MOST COMMON STALL

Adding the bottoms: $1/2 + 1/4 = 2/6$. Go back to the bar picture, not to more practice.

Mid-Unit Quiz

8 items • Weeks 1–3 • 7 of 8 keeps you flying

____ / 8

1. $\frac{1}{2} = \underline{\hspace{1cm}} / 8$
.....

2. Write $\frac{10}{15}$ in simplest form.
.....3. Mark $\frac{3}{4}$ and $\frac{5}{8}$ on this line. Which is bigger?

0 _____ 1

4. Which is bigger, $\frac{2}{3}$ or $\frac{3}{5}$? Show both in fifteenths.
.....

5. $\frac{3}{8} + \frac{2}{8} = \underline{\hspace{1cm}}$
.....

6. $\frac{1}{2} + \frac{1}{4} = \underline{\hspace{1cm}}$
.....

7. $\frac{2}{5}$ of 30 = $\underline{\hspace{1cm}}$
.....

CHALLENGE ITEM

8. $\frac{3}{4} = \square / 12$ $\square = \underline{\hspace{1cm}}$

What the bottom was multiplied by:

.....

Mission 04 Test

12 items + one explanation • part 1 of 2

1. $\frac{1}{4} = \frac{\quad}{12}$

.....

2. $\frac{6}{8}$ in simplest form

.....

3. How many sixths in 2 wholes?

.....

4. Mark $\frac{2}{3}$ and $\frac{5}{6}$ on this line, then write which is bigger and by how much.

0 _____ 1

5. Which is bigger, $\frac{5}{6}$ or $\frac{7}{9}$? Show both in eighteenths.

.....

6. Order smallest first: $\frac{3}{5}$, $\frac{2}{3}$, $\frac{7}{10}$.

.....

7. $\frac{5}{6} - \frac{2}{6}$

.....

8. $\frac{2}{3} + \frac{1}{6}$

.....

9. $\frac{3}{4} - \frac{1}{8}$

.....

10. $2 - \frac{3}{5} = \frac{\quad}{\quad}$. Show how you turned the 2 into fifths.

.....

Reasoning & the Big Question

11. $\frac{3}{8}$ of 40 = _____

.....

12. $\frac{5}{6}$ of 42 = _____

.....

CHALLENGE ITEM

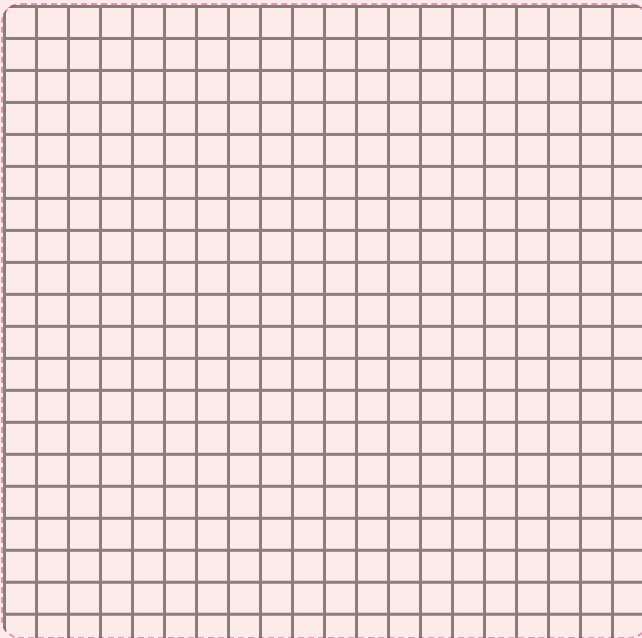
$\frac{3}{4}$ of a number is 18. What is the number? Show the step that got you there.

.....

EXPLAIN YOUR THINKING • WORTH 4 POINTS

How can two different-looking fractions be exactly the same number?

Take $\frac{2}{3}$ and $\frac{8}{12}$. Draw both on the same bar and show they cover the same amount. Then say what changed and what didn't. Spelling doesn't count. Reasoning does.



.....

.....

.....

.....

.....

.....

.....

TROPHY BAND • CIRCLE ONE

85–100%

Mission Commander

Equivalence Badge earned.
Mission 05 opens — and
Challenge becomes your Core.

70–84%

Flight Engineer

Two targeted days on the
flagged items, then retake those
items only.

Below 70%

Ground Crew

Back to paper strips and the bar
picture before any more symbol
work.

Key A • Pre-Assessment, Lessons 1.1–1.2

Pre-Assessment

A1 mark three-quarters of the way along

A2 6 thirds

B1 $\frac{4}{8}$ B2 $\frac{9}{12}$

C1 $\frac{2}{3}$ (10/15 against 9/15)

C2 $\frac{2}{5}$, $\frac{1}{2}$, $\frac{3}{4}$

D1 $\frac{5}{8}$ D2 $\frac{3}{4}$

E1 $\frac{2}{3}$ E2 $\frac{3}{4}$

F1 12 F2 24

Scoring note. D2 answered as $\frac{2}{6}$ is the single most diagnostic error in this mission. Mark strand D new and run Week 4 in full, whatever else he scored.

Lesson 1.1 • One Whole, Cut Up

Warm-Up 2 • 4 • 8 • 2 • 4 • 1

Bar rows $\frac{1}{2}$ • $\frac{1}{2}$ • $\frac{2}{4}$ • $\frac{4}{8}$

Core

1. $\frac{3}{4}$ 2. 6 thirds

3. 1 whole 4. 1 whole and 3 quarters

5. $\frac{5}{8}$ sits between $\frac{1}{2}$ and $\frac{3}{4}$

Lesson 1.1 Challenge. C1 • four marks. C2 • two eighths apart. C3 • $\frac{1}{5}$ is bigger; cutting the same whole into more pieces makes each piece smaller, so a bigger bottom number means a smaller piece. Accept the bar drawing as the whole answer.

Lesson 1.2 • Same Number, New Name

Warm-Up 3 • 2 • 4 • 2 • 6 • 2

Core (multiplier • answer)

1. $\times 3 \rightarrow \frac{9}{12}$

2. $\times 4 \rightarrow \frac{8}{20}$

3. $\div 2 \rightarrow \frac{3}{4}$

4. $\div 5 \rightarrow \frac{2}{3}$

5. $\div 3 \rightarrow \frac{3}{4}$

Challenge. C1 • $\frac{2}{3}$; both divided by 4. C2 • $\frac{2}{4}$, $\frac{3}{6}$, $\frac{4}{8}$, $\frac{5}{10}$, $\frac{6}{12}$, $\frac{7}{14}$, $\frac{8}{16}$, $\frac{9}{18}$ — with $\frac{1}{2}$ itself that is **nine** fractions; every one has a top exactly half its bottom. C3 • any drawing where a bar in thirds is re-cut into twelfths and the shaded part is unchanged. The point is that nothing was added — the pieces were only subdivided.

Watch for: multiplying only the bottom. Ask “what did you do to the pieces?” — if every piece was cut in two, there must now be twice as many shaded ones too.

Key B • Lessons 1.3–1.4, Enrichment, Quiz

Lesson 1.3 • Which Is Bigger

Warm-Up $1/2 \cdot 1/3 \cdot 3/4 \cdot 3/5 \cdot 5/6 \cdot 7/10$

Core

1. $10/15$ and $9/15 \rightarrow 2/3$
2. $6/8$ and $5/8 \rightarrow 3/4$
3. $15/18$ and $14/18 \rightarrow 5/6$
4. $8/12$ and $9/12 \rightarrow 3/4$
5. $7/8$ — a gap of $1/8$ is smaller than a gap of $1/6$

Challenge. C1 $\cdot 18/30, 20/30, 21/30 \rightarrow 3/5 < 2/3 < 7/10$.
C2 $\cdot 5/8$. C3 $\cdot$ less than a half, because half of 15 is 7.5 and the top is only 7. Any “compare the top to half the bottom” reasoning earns full credit.

Lesson 1.4 • Adding

Warm-Up $2/4 \cdot 3/5 \cdot 5/8 \cdot 3/6 \cdot 3/10 \cdot 1 \text{ whole}$

1. $2/4 + 1/4 = 3/4$
2. $4/6 + 1/6 = 5/6$
3. $6/8 - 1/8 = 5/8$
4. $5/6 - 2/6 = 3/6 = 1/2$
5. $3/6 + 2/6 = 5/6$ — both become sixths

Challenge. C1 $\cdot 9/12 + 8/12 = 17/12$, or 1 and $5/12$. He should spot it exceeds 1 because $3/4$ alone is more than a half and so is $2/3$. C2 $\cdot 10/5 - 3/5 = 7/5 = 1$ and $2/5$. C3 $\cdot$ they added both tops and both bottoms. The bar shows $1/2 + 1/4$ covering three quarters of the strip, while $2/6$ is less than half — so the answer got smaller than one of the things being added, which is impossible.

Friday • Fold the Universe

Shared creases. $1/2 = 2/4 = 4/8 = 8/16$; $1/4 = 2/8 = 4/16$; $3/4 = 6/8 = 12/16$. Any three correct pairs count.

Fraction of a set. $5 \cdot 12 \cdot 15 \cdot 30 \cdot 35 \cdot 24$

The real question. With a 5 and a 6, build $5/6$ — it's just under 1, and $6/5$ goes over and scores nothing. The best possible hand is any pair of consecutive numbers with the smaller on top: $9/10$ is the closest to 1 you can get from a stripped deck, at one tenth away.

Mid-Unit Quiz • out of 8

1. $4/8$
2. $2/3$
3. $3/4$ is bigger ($6/8$ against $5/8$)
4. $2/3 - 10/15$ against $9/15$
5. $5/8$
6. $3/4$
7. 12
8. $\square = 9$, bottom $\times 3$

7/8 or better $\rightarrow$ continue. A miss on 6 means Week 4 needs the bar picture on day one. A miss on 3 means the number line never really landed — go back to the folded strip before Week 4, not after.

Key C · Mission 04 Test & Scoring

Items 1–12 · one point each

- 1. 3/12
- 2. 3/4
- 3. 12 sixths
- 4. 5/6 is bigger, by 1/6 (4/6 against 5/6)
- 5. 15/18 and 14/18 → 5/6
- 6. 3/5, 2/3, 7/10
- 7. 3/6 = 1/2
- 8. 4/6 + 1/6 = 5/6
- 9. 6/8 - 1/8 = 5/8
- 10. 10/5 - 3/5 = 7/5 (1 and 2/5)
- 11. 15
- 12. 35

Challenge item. $18 \div 3 = 6$ is one quarter, so the number is **24**.

Explanation · 4 points

- Score the reasoning, not the writing. Award one point each:
- ☐ 1 · Draws both fractions on the same-sized bar
 - ☐ 2 · Shows the shaded amounts matching exactly
 - ☐ 3 · Names what changed: the number and size of the pieces
 - ☐ 4 · Names what didn't: the amount of bar covered

Model answer, kid-sized. “2/3 and 8/12 shade the same part of the bar. I just cut each third into four little bits, so there are more pieces and each one is smaller. Nothing was added or taken away — only the pieces changed, not the amount.”

Total 16 points. 14+ Mission Commander · 11–13 Flight Engineer · 10 or fewer Ground Crew. Items 8, 9 and 10 missed together mean unlike denominators, not fractions in general — two targeted days on Week 4 and retake those three.

Error Journal · photocopy this page

One row per mistake, for the whole mission. The “why” line must name a step. “I was careless” doesn't count.

DATE	PROBLEM	I GOT / ANSWER	WHY IT HAPPENED — ONE SENTENCE